Teoría

$$\begin{cases} \cosh(s) = 1 + \frac{1}{2!}s^2 + \frac{1}{4!}s^4 + \frac{1}{6!}s^6 + \cdots \\ \operatorname{senh}(s) = s + \frac{1}{3!}s^3 + \frac{1}{5!}s^5 + \frac{1}{7!}s^7 + \cdots \end{cases}$$

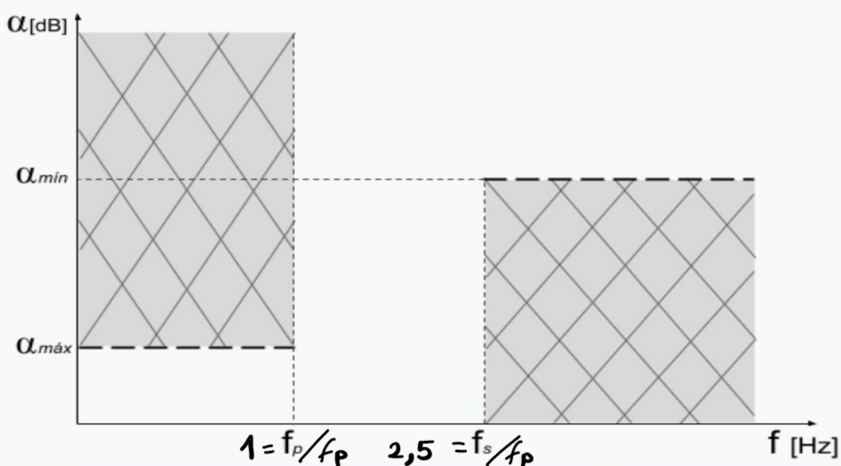
$$\circ\ \operatorname{Cotgh}(s) = \frac{\cosh(s)}{\operatorname{senh}(s)} \longrightarrow \boxed{\operatorname{cotgh}(s) = \frac{1}{s} + \cfrac{1}{\frac{3}{s} + \cfrac{1}{\frac{5}{s} + \cfrac{1}{\frac{7}{s} + \cdots}}}}$$

$$\frac{\sigma_j}{\sigma_j^2 + (\omega - \omega_j)^2}$$

$$D(\omega) = \sum_j \left(\frac{\sigma_j}{\sigma_j^2 + (\omega - \omega_j)^2} \right) - \sum_i \left(\frac{\sigma_i}{\sigma_i^2 + (\omega - \omega_i)^2} \right)$$

$$\begin{cases} B(s) = \operatorname{Num}(\operatorname{cotgh}(s)) + \operatorname{Den}(\operatorname{cotgh}(s)) \\ B_n(s) = (2n - 1)\, B_{n-1} + s^2 \cdot B_{n-2} \\ T_n(s) = \dfrac{B_n(0)}{B_n(s)} \end{cases}$$

$$\boxed{\xi^2 = 10^{\left(\frac{\alpha_{max}}{10}\right)} - 1} \qquad \boxed{|\alpha|_{dB} = 10 \cdot \operatorname{Log}(1 + \xi^2 \cdot \omega^{2n})}$$



D(w) [seg]	αmáx [dB]	wp [rad/seg]	ws [rad/seg]	αmín [dB]
1	1	1	2.5	5

1)

n = 2 :

$$\operatorname{cotgh}_2(s) \cong \frac{1}{s} + \frac{s}{3} = \frac{s^2+3}{3s} \quad //$$

$$\begin{cases} B_2(s) = s^2 + 3s + 3 \\ B_2(0) = 3 \end{cases}$$

$$\Longrightarrow T_2(s) = \frac{3}{s^2+3s+3} \quad //$$

n = 3 :

$$\operatorname{cotgh}_3(s) \cong \frac{1}{s} + \frac{1}{\frac{3}{s} + \frac{s}{5}} = \frac{6s^2+15}{s^3+15s}$$

$$\begin{cases} B_3(s) = s^3 + 6s^2 + 15s + 15 \\ B_3(0) = 15 \end{cases}$$

$$\Longrightarrow T_3(s) = \frac{15}{s^3+6s^2+15s+15} \quad //$$

n = 4 :

$$\operatorname{cotgh}_4(s) \cong \frac{1}{s} + \frac{1}{\frac{3}{s} + \frac{1}{\frac{5}{s} + \frac{s}{7}}} = \frac{1}{s} + \frac{(s^2+35)s}{10s^2+105} = \frac{10s^2+105+s^4+35s^2}{10s^3+105s}$$

$$\begin{cases} B_4(s) = s^4 + 10s^3 + 45s^2 + 105s + 105 \\ B_4(0) = 105 \end{cases}$$

$$\Longrightarrow T_4(s) = \frac{105}{s^4+10s^3+45s^2+105s+105} \quad //$$

2)

- $\varepsilon^2 = 10^{\alpha_{max\,dB}/10} - 1 \longrightarrow \varepsilon^2 = 0,2589$

$$|\alpha|\Big/_{\alpha=\alpha_{min}} = \alpha_{min} = 10\,\mathrm{Log}\left(1 + \varepsilon^2 \cdot (f_s/f_p)^{2n}\right)$$

n	α_{min} [dB]
1	4,18
2	10,46 > 5

$$\Longrightarrow \begin{cases} \varepsilon^2 = 0,2589 \\ n = 2 \end{cases}$$

$$\Longrightarrow T_2(s) = \frac{3}{s^2+3s+3} \;//$$

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3)

- $T(s) = \dfrac{3}{s^2+3s+3} = \dfrac{3}{\left(-\frac{3}{2} \pm j\frac{\sqrt{3}}{2}\right)}$

- $D(\omega) = \sum_j \left(\dfrac{\sigma_j}{\sigma_j^2 + (\omega - \omega_j)^2} \right) \Big/$ "Ya que la $T(s)$ no contiene ceros"

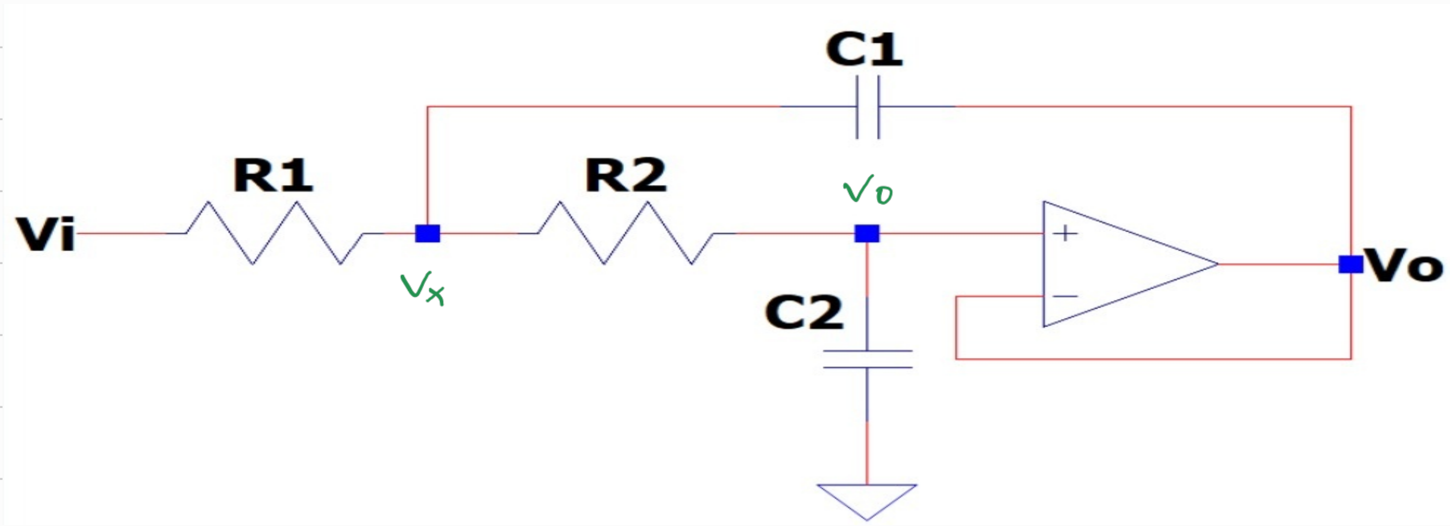
$$D(\omega) = \frac{\sigma_{p1}}{\sigma_{p1}^2 + (\omega - \omega_{p1})^2} + \frac{\sigma_{p2}}{\sigma_{p2}^2 + (\omega - \omega_{p2})^2}$$

$$D(\omega) = \frac{3/2}{9/4 + (\omega - \sqrt{3}/2)^2} + \frac{3/2}{9/4 + (\omega + \sqrt{3}/2)^2}$$

$$\begin{cases} D(0) = 1 \\ D(2,5) = 0,415 \end{cases}$$

$$\Longrightarrow \Delta D_{\%} = (D(0) - D(2,5)) \cdot 100 = 58,4\% \;//$$

4) Sallen - Key



$$\begin{cases} 0 = (V_x - V_i)\,G_1 + (V_x - V_o)\,G_2 + (V_x - V_o)\,s\,C_1 \\ (V_x - V_o)\,G_2 = V_o\,s\,C_2 \end{cases}$$

$$\Longrightarrow T(s) = \frac{G_1 G_2}{C_1 C_2\, s^2 + C_2 G_1\, s + C_2 G_2\, s + G_1 G_2}$$

$$T(s) = \frac{\frac{G_1 G_2}{C_1 C_2}}{s^2 + s\,\frac{G_1 + G_2}{C_1} + \frac{G_1 G_2}{C_1 C_2}}$$

$$\boxed{T(s) = \frac{\frac{1}{C_1 C_2 R_1 R_2}}{s^2 + s \cdot \frac{R_1 + R_2}{C_1 R_1 R_2} + \frac{1}{C_1 C_2 R_1 R_2}}}$$

$$\boxed{\omega_0 = \frac{1}{\sqrt{C_1 C_2 R_1 R_2}}}$$

$$\frac{\omega_0}{Q} = \frac{R_1 + R_2}{C_1 R_1 R_2}$$

$$\hookrightarrow Q = \frac{C_1 R_1 R_2\, \omega_0}{R_1 + R_2} \rightarrow \boxed{Q = \sqrt{\frac{C_1}{C_2}} \cdot \frac{\sqrt{R_1 R_2}}{R_1 + R_2}}$$

$$\bullet\ T_{BT}(s)\Big/_{n=2} = \frac{3}{s^2 + 3s + 3}$$

$$\Rightarrow \begin{cases} \sqrt{3} = \dfrac{1}{\sqrt{C_1 C_2 R_1 R_2}} & \longrightarrow\ C_1 = \dfrac{1}{3 C_2 R_1 R_2} \\[2ex] \dfrac{\sqrt{3}}{3} = \sqrt{\dfrac{C_1}{C_2}} \cdot \dfrac{\sqrt{R_1 R_2}}{R_1 + R_2} & \longrightarrow\ \dfrac{\sqrt{3}}{3} = \sqrt{\dfrac{1}{3 C_2^2 \cancel{R_1 R_2}}} \cdot \dfrac{\cancel{\sqrt{R_1 R_2}}}{R_1 + R_2} \end{cases}$$

$$\Rightarrow C_2 = \frac{1}{R_1 + R_2}$$

$$\Longrightarrow \boxed{\begin{aligned} C_1 &= \frac{1}{3(R_1 /\!/ R_2)} \\ C_2 &= \frac{1}{R_1 + R_2} \end{aligned}}$$

$\Longrightarrow$ • Si $C_1 = 10\,\mu F \wedge C_2 = 100\,nF$

$\Longrightarrow R_1 = 10\,M\Omega \wedge R_2 = 33{,}4\,K\Omega$